

AP CALCULUS AB - UNIT 1

Guided Notes 1.3: Estimating Limit Values from Graphs



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we trace a graph toward a target from each side before reading any point marker?

- Trace a graph toward a target from each side before reading any point marker.
- Estimate one-sided and two-sided limits from graphical evidence.
- Distinguish holes, jumps, vertical asymptotes, endpoints, corners, and oscillation.
- State the precision supported by the scale and viewing window.

Key Knowledge, Vocabulary, and Notation

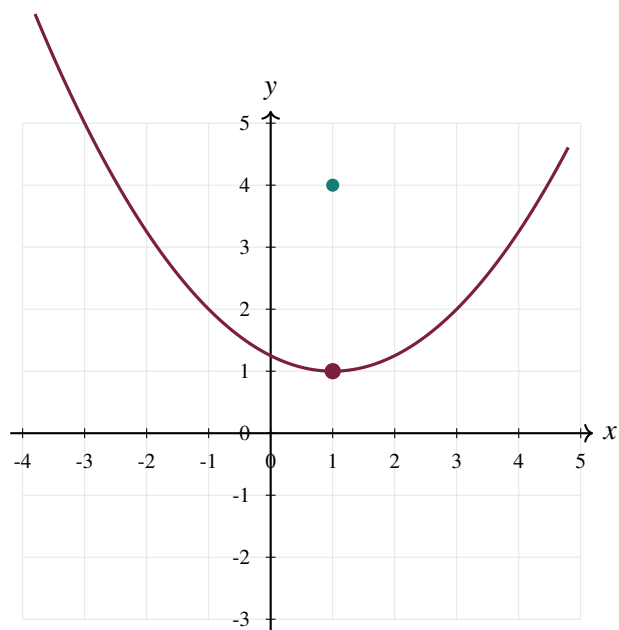
Trace, do not land	Follow the curve toward the target; read a filled point only when asked for the function value.
Left/right evidence	A two-sided limit requires the same vertical destination from both directions.
Endpoint rule	At a domain endpoint, only the admissible one-sided limit is relevant.
Unbounded behavior	Record the sign of each branch using $+\infty$ or $-\infty$.
Graph precision	A graph supports an estimate, not arbitrary decimals beyond its scale.

Concept Development and Strategy

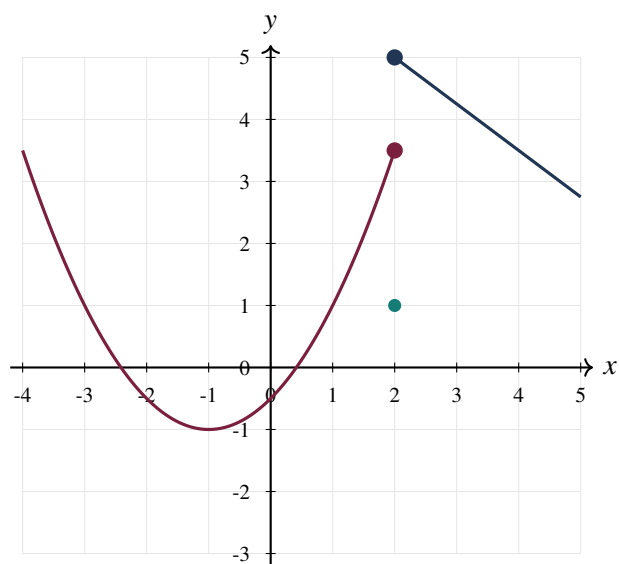
1. Cover the filled point with your finger and trace the curve from the left and right.
2. Write the two one-sided limits before deciding the two-sided limit.
3. For a vertical asymptote, inspect branch direction rather than merely naming the vertical line.

Shared Representation / Data

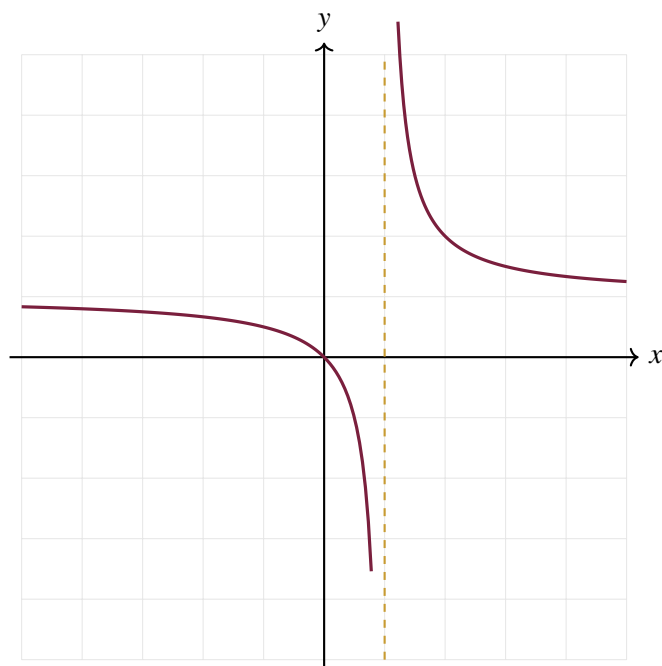
Graph A



Graph B



Graph C


Worked Example: Hole with a different point value

In Graph A, determine $\lim_{x \rightarrow 1} f(x)$ and $f(1)$.

Answer and Reasoning

The curve approaches the open point $(1, 1)$ from both sides, so the limit is 1. The filled point is at $(1, 4)$, so $f(1) = 4$.

Worked Example: Jump behavior

In Graph B, determine the left-hand and right-hand limits at $x = 2$.

Answer and Reasoning

The left branch approaches 3.5 and the right branch approaches 5. Since they differ, the two-sided limit does not exist. The filled point gives $f(2) = 1$ but does not affect the limit.

Worked Example: Vertical asymptote signs

In Graph C, describe the behavior at $x = 1$.

Answer and Reasoning

From the left, the branch decreases without bound: $\lim_{x \rightarrow 1^-} f(x) = -\infty$. From the right, it increases without bound: $\lim_{x \rightarrow 1^+} f(x) = +\infty$.

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. Using Graph A, find $\lim_{x \rightarrow 1^-} f(x)$.

2. Using Graph A, find $f(1)$.

3. Using Graph A, explain why changing the filled point would not change the limit at $x = 1$.

4. Using Graph B, state $\lim_{x \rightarrow 2^-} f(x)$ and $\lim_{x \rightarrow 2^+} f(x)$.

5. A graphing window shows a curve approaching about 2.4 near $x = 5$, with vertical grid spacing 1. Which is more defensible: 2.4 or 2.41763? Explain.

6. Construct a verbal description of a graph satisfying $\lim_{x \rightarrow 2^-} f(x) = -1$, $\lim_{x \rightarrow 2^+} f(x) = 4$, and $f(2) = 0$.

7. Draw a graph with $f(0) = 2$, $\lim_{x \rightarrow 0} f(x) = -1$, and a vertical asymptote at $x = 3$ with both sides tending to $+\infty$. Label all special points.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

- At $x = a$, a graph approaches 6 from both sides but has a filled point at height -3 . Which statement is true?
(A) $\lim_{x \rightarrow a} f(x) = -3$ (B) $\lim_{x \rightarrow a} f(x) = 6$
(C) The limit DNE. (D) $f(a) = 6$
- Which graphical feature necessarily prevents a finite two-sided limit at the target?
(A) A corner (B) A cusp whose sides meet
(C) A jump with unequal one-sided heights (D) A hole
- A calculator graph suggests a limit of 1.732 at $x = 2$, but the curve is steep and the window uses y -scale 10. What is the best next step?
(A) Report 1.732000 immediately. (B) Zoom and confirm with a numerical table from both sides.
(C) Read the filled point only. (D) Conclude DNE.

Common Errors and AP Precision

- Keep the value of a function at a point separate from its limiting behavior near that point.
- State one-sided behavior explicitly whenever a two-sided limit or continuity claim depends on agreement from both sides.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

Can a graph cross a horizontal line $y = L$ infinitely many times near $x = c$ and still satisfy $\lim_{x \rightarrow c} f(x) = L$? Explain.

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